



Subseasonal prediction of early summer precipitation in the middle and lower reaches of the Yangtze River Basin based on circulation classification

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ABSTRACT

Employing a Self-Organizing Map (SOM)-based circulation classification technique, this study establishes a nonlinear relationship between circulation patterns and daily precipitation probability distribution during early summer (1961–1990) in the middle and lower reaches of the Yangtze River Basin (MLYRB). Based on this relationship, statistical downscaling models were developed at monthly and pentad scales to capture refined spatial structure of subseasonal precipitation. Following independent tests from 1991 to 2002, the models were driven by daily outputs from the NCEP CFSv2 climate model during 2015–2022 to assess prediction skill. Evaluation results indicate that the proportion of years with statistically significant spatial correlation coefficients between observed and predicted precipitation increased from 60 % (53 %) before downscaling to 95 % (92 %) after downscaling for monthly (pentad) predictions. The multi-year mean pattern correlation coefficients nearly doubled, rising by approximately 0.3 and 0.2 for monthly and pentad scales, respectively. Substantial improvements are also found in the normalized standard deviations and centered root mean square errors, indicating robust long-term predictive performance. Furthermore, the downscaling approach promotes consistency across lead times within two weeks for monthly precipitation predictions (within one month for pentad predictions), demonstrating extended predictable lead times. Overall, the SOM-based statistical downscaling models significantly enhance both the accuracy and temporal predictability of subseasonal precipitation predictions during early summer in MLYRB, showing strong potential for operational regional climate prediction.

1. Introduction

The middle and lower reaches of the Yangtze River Basin (MLYRB), one of the most economically developed and densely populated regions in China, have experienced frequent and intense floods and droughts in history (Dou et al., 2020; Zhou et al., 2020; Ding et al., 2021; Liu et al., 2021b). The major rainy season over the MLYRB, known as Meiyu in China, is caused by a quasi-stationary rainbelt in early summer, which typically brings prolonged rainfall, reduced solar radiation, and elevated soil moisture levels. These conditions create cumulative and delayed hazards, particularly affecting agricultural productivity, watershed safety, and the ecological environment. Thus, an accurate forecast of

early summer rainfall over the MLYRB is of great significance for disaster prevention and mitigation. In recent years, advances in dynamic numerical forecasting and the expansion of observational datasets have significantly improved the accuracy of medium- and short-term deterministic forecasts (ranging from several hours to two weeks), as well as probabilistic forecasts on seasonal scales (three to six months). However, the factors influencing subseasonal (15–60 days) predictions are complex, including sea surface temperatures, atmospheric internal dynamical processes which are unpredictable components of the climate system (He et al., 2016; Kosaka et al., 2012), as well as land surface processes and stratospheric processes (Xiao and Duan, 2016; Zhao et al., 2018). Thus, predictability at the subseasonal scale remains notably

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limited, leading to the improvement of the prediction capabilities on this scale becoming a key yet difficult frontier in climate research (Vitar et al., 2017; Kumar and Chen, 2017; He, 2020; Liu et al., 2021a; Wang et al., 2021).

Dynamic climate models serve as powerful tools for predicting regional subseasonal precipitation. Operational climate prediction systems developed by major meteorological centers worldwide (e.g., CMA_CPSv3, ECMWF_SYSS, NCEP_CFSv2, and TCC_CPS3) are capable of reasonably predicting some fundamental climate characteristics of summer precipitation over China, including the spatial pattern of summer mean precipitation and the standard deviation of summer precipitation (Liu et al., 2019). However, these models exhibit considerable biases in simulating the variance of precipitation and its subseasonal temporal variability (Huang et al., 2015; Chen et al., 2019; Liu et al., 2021a). Furthermore, prediction skill declines markedly as lead time increases, with the skill for pentad mean precipitation over the MLYRB during summer limited to approximately one week (Liu et al., 2021a), and the predictive capability of monthly precipitation over the eastern China only covers 5–11 days (Zhang et al., 2019). Despite these limitations in precipitation forecasting, the models demonstrate relatively strong performance in simulating large-scale circulation patterns that influence the spatial distribution of precipitation. Furthermore, the predictable lead times of these circulation features tend to exceed that of precipitation itself (Ding, 2011; Huang et al., 2015; Zhao et al., 2019; Wu et al., 2023; Zhu et al., 2024). This discrepancy may be attributed to the models' limited capacity to accurately represent the complex relationships between regional precipitation and its influencing factors (Kosaka et al., 2012; Liu et al., 2015; Liu et al., 2021a). Therefore, if we could identify robust empirical relationships between configurations of key influencing factors and precipitation patterns by leveraging the skillful outputs of dynamic models and observational data, it would be possible to correct systematic model errors and enhance the accuracy of subseasonal precipitation forecasts. Such improvements can be achieved through statistical prediction methods grounded in model interpretation frameworks (Jia et al., 2013; Feng et al., 2013).

Current statistical approaches applied in subseasonal precipitation prediction include both relatively mature statistical downscaling methods and emerging artificial intelligence (AI)-based techniques (Yang et al., 2022). Statistical downscaling methods typically involve establishing empirical relationships between predictors and predictands. Techniques such as Empirical Orthogonal Function (EOF), Canonical Correlation Analysis (CCA), and Singular Value Decomposition (SVD) can effectively enhance prediction skills. For instance, over the coastal region of South China, an approach combining the SVD method with a dynamical model has been shown to improve both the temporal variability and spatial distribution of winter (summer) precipitation at lead times beyond 15–20 days (Zhang et al., 2023). Similarly, for monthly precipitation over the Huaihe River Basin, a hybrid spatio-temporal statistical downscaling model incorporating EOF, CCA, and other statistical methods has demonstrated the ability to reduce prediction errors. Nonetheless, during June in the 2015–2017 period, only the downscaling results for 2017 outperformed the dynamical model (Liu et al., 2018). It is noted that some traditional statistical downscaling approaches which rely on transformation functions generally establish primary relationships between monthly (or seasonal) averages of large-scale climate fields and local precipitation based on statistical indicators such as variance contribution. This limits their ability to depict the one-to-one physical relationships among all the circulation and precipitation samples over sub-monthly timescales (Wilby et al., 2004). As for artificial intelligence techniques, they can offer notable advantages in modeling complex daily nonlinear relationships (Weyn et al., 2021; Wang et al., 2021; Li et al., 2023; Chen et al., 2023; Ye et al., 2024). But their practical application is currently constrained by the need for large training datasets, high computational costs, and a susceptibility to overfitting, which can diminish their generalizability and independent forecasting skill. As such, there

remains considerable scope for further optimization and refinement of AI-based predictive models.

As a weather classification technique based on an unsupervised artificial neural network, the Self-Organizing Map (SOM) has been widely recognized for its ability to integrate the strengths of both statistical downscaling and machine learning. Numerous studies have demonstrated that SOM can efficiently establish one-to-one nonlinear physical relationships between daily circulation patterns and the probability distribution of observed precipitation (Cavazos, 1999; Hewitson and Crane, 2002; Wilby et al., 2004; Fowler et al., 2007; Yin et al., 2011). Originally proposed by Kohonen (1982, 1990), the SOM method was introduced into atmospheric science by Hewitson and Crane (2002) to establish the nonlinear association between circulation and local precipitation probability distributions, enabling the identification of key circulation patterns influencing extreme precipitation events. Hewitson and Crane (2006) further applied the SOM method to statistical downscaling, conducting experiments in South Africa that demonstrated its effectiveness in capturing the statistical characteristics of daily precipitation based on the relationship between circulation pattern and precipitation distribution. Previous researches have shown that the iterative training process inherent to SOM enhances simulation variance and better captures the nonlinear dynamics of atmospheric circulation, thereby providing a more comprehensive representation of circulation–precipitation relationships (Cavazos, 1999; Hewitson and Crane, 2002; Yin et al., 2011). Additionally, the unsupervised learning framework of SOM offers an objective classification advantage, while its two-layer neural network structure reduces computational demand and improves interpretability compared to deeper learning models (Wilby et al., 2004; Shen et al., 2020; Li et al., 2020). Applied in conjunction with global climate models, the SOM approach has been successfully used for long-term downscaling studies in regions including Australia, the United States, and China, with its effectiveness validated in these contexts (Yin et al., 2011; Ning et al., 2012; Li et al., 2020). However, it is important to note that climate models for long-term projection primarily account for externally forced responses, whereas subseasonal prediction models must incorporate both initial atmospheric conditions and slowly varying factors such as sea surface temperature. Consequently, the applicability and effectiveness of the SOM-based downscaling approach at the subseasonal scale remain to be systematically evaluated.

This study aims to integrate the SOM-based circulation classification technique to identify key circulation patterns influencing subseasonal precipitation in early summer over MLYRB. By establishing nonlinear relationships between these circulation patterns and the probability distribution of daily precipitation, statistical downscaling models are developed to capture the fine-scale spatial structure of subseasonal precipitation in the region. The performance of this downscaling approach is then evaluated in combination with outputs from the subseasonal dynamic climate model. The structure of the paper is as follows: Section 2 introduces the data and methodology. Section 3 presents the construction of the subseasonal downscaling prediction model for MLYRB based on the SOM method, with a focus on the monthly and pentad scales. Section 4 evaluates the predictive skill improvements achieved by the downscaling models. Section 5 concludes with a summary and discussion of the findings.

2. Data and methods

2.1. Data

The observational dataset used in this study comprises daily precipitation records at more than 2400 meteorological stations in China from 1961 to 2002 and 2015–2022, covering early summer (June), provided by the China Meteorological Administration. A total of 56 stations located in MLYRB (109°–124°E, 27°–32.5°N) are selected for analysis (Fig. 1). The large-scale circulation data used for classification

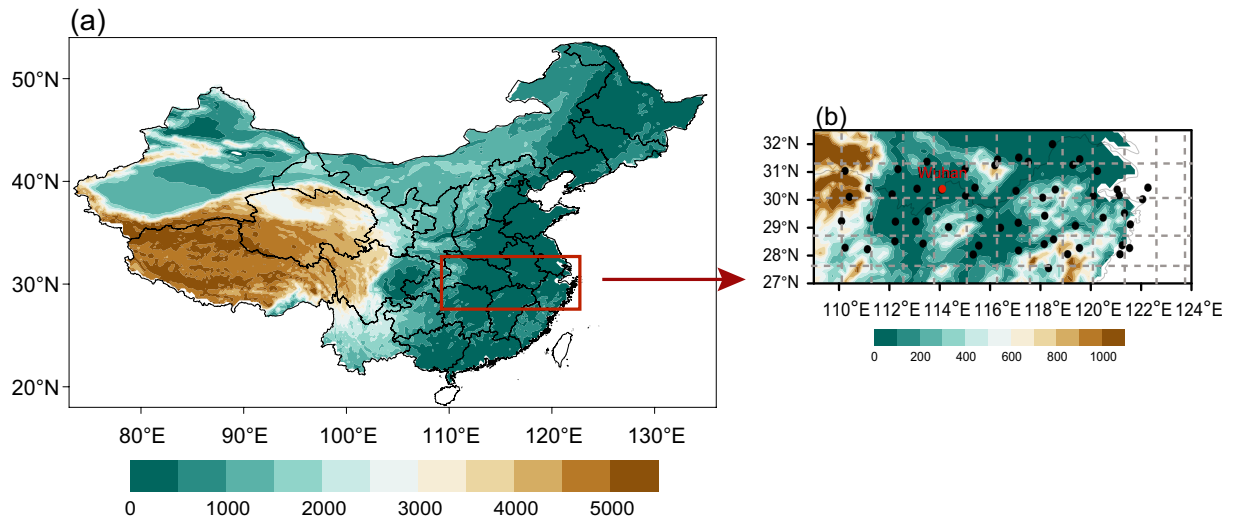


Fig. 1. (a) The location of the middle and lower reaches of the Yangtze River Basin (MLYRB) within China (the red box). (b) Distribution of the 56 stations in the MLYRB. The shading indicates the terrain height, and the gray dashed lines represent the unified grids of the reanalysis data and dynamic model datasets. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

are derived from ERA5 daily reanalysis data from the European Centre for Medium-Range Weather Forecasts (ECMWF), spanning the period 1961–2002 with a spatial resolution of $0.25^\circ \times 0.25^\circ$ (Hersbach et al., 2019). For model development, observational and reanalysis data from 1961 to 1990 are used to train the statistical downscaling models, while data from 1991 to 2002 are reserved for independent model validation.

To assess the predictive skill improvements achieved by the SOM downscaling approach, circulation data from the dynamic climate models are utilized to drive the downscaling models for the period 2015–2022. Ensemble means of hindcasts and realtime predictions initiated from January 1982 to December 2022 from the NCEP Climate Forecast System Version 2 (CFSv2), which is often used in the sub-seasonal climate prediction operation, are applied in this work (Xue et al., 2013; Saha et al., 2014). The ensemble means of the realtime predictions include 16 forecast members from the initial date, out to 45 days, with the spatial resolution of $1.5^\circ \times 1.5^\circ$. In order to project the circulation data from CFSv2 onto the circulation patterns in the training period, the spatial resolutions of the ERA5 reanalysis data and the datasets from CFSv2 model are unified to $1.25^\circ \times 1.25^\circ$. The downscaled precipitation outputs are then compared with precipitation from CFSv2 model and corresponding observational data to conduct a comprehensive performance evaluation.

2.2. Statistical downscaling framework based on circulation classification

The Self-Organizing Map (SOM) is a type of neural network model designed to simulate the lateral interactions among neurons (Kohonen, 1982). It employs an unsupervised learning algorithm, where the fundamental principle involves competitive learning: each neuron in the output layer competes to best match the input pattern, and the winning neuron represents the cluster to which the input is assigned (Kohonen, 1990). The core SOM algorithm has been extensively described in prior studies (Li et al., 2020; Li et al., 2024). In this study, the daily fields of circulation variables serve as the input to the SOM, while the output consists of the key circulation patterns identified through clustering. The selection of circulation factors is based on the assessment of atmospheric systems that exert significant influence on early summer precipitation in MLYRB. These include the Western Pacific Subtropical High (WPSH) (Wang and Wu, 1996; Tao et al., 2001; Guan et al., 2019), the East Asian westerly trough (Zhang et al., 2005; Li et al., 2014), and the blocking high (Li, 2005; Shi and Zhi, 2007). Taking into account the active regions of these systems during early summer, the 500 hPa geopotential

height field over the domain $40^\circ\text{--}160^\circ\text{E}$ and $0^\circ\text{--}70^\circ\text{N}$ is selected as the circulation variable for SOM classification. Based on this input, statistical downscaling models are constructed at both monthly and pentad scales. To filter out high-frequency weather-scale noise while preserving lower-frequency climate-scale signals, a 5-day moving average is applied to the geopotential height data. For the monthly-scale model, circulation and precipitation information for June, along with the 15 days before and after, are used to capture a broader range of circulation–precipitation relationships that may arise during early summer. For the pentad-scale model, each pentad period is modeled using the corresponding circulation and precipitation data, including the 5 days preceding and following each pentad. In this study, the training period, independent testing period, and evaluation period accounts for 60 %, 24 %, and 16 % of the total duration, respectively. The specific procedures for model construction and evaluation are outlined as follows:

Training Period (1961–1990): Prior to performing circulation classification, some input parameters are needed to be determined. Previous researches indicates that, given specific cluster number and size, classification outcomes are not highly sensitive to the selection of other parameters (Harrington et al., 2016; Tan et al., 2020). Thus, we evaluate the classification quality with different number and size of circulation patterns. In this study, two metrics are employed to assess classification performance: the quantization error (QE), represented by the average Euclidean distance between the samples belonging to a cluster and the reference vector of the best-matching neurons, which reflects the ability of each best-matching neuron representing intra-cluster samples; and the total quantization error (TQE), which accounts for both the individual cluster errors and the number of classifications (Liu et al., 2016). The two indicators are calculated as (both of them are standardized):

$$QE = \frac{1}{n} \sum_{i=1}^n \|\vec{x}_i - m_{\vec{x}_i}\| \quad (1)$$

$$TQE = QE \times N \quad (2)$$

where, $m_{\vec{x}_i}$ is the best matching prototype of the corresponding data-vector $\vec{x}_i$, n is the number of data-vectors, and N represents the number of patterns.

Classification quality are considered optimal when both error measures are relatively low. Notably, the selection of the SOM classification number and size should consider not only classification quality but also the ability to effectively depict climate features after clustering. This ensures an adequate classification number to represent primary patterns

of impact factor without excessive patterns that might obscure the characteristic information of factor (Li et al., 2024). After comparing multiple classification configurations, a 12 patterns (4 rows by 3 columns) is selected for the monthly scale, and a 4 patterns (4 rows by 1 columns) configuration is chosen for the pentad classification (Fig. 2). Based on these sizes, key circulation patterns are identified through training. Subsequently, the empirical relationship between each circulation pattern and the observed daily precipitation probability distribution in MLYRB is established.

Independent Testing Period (1991–2002): An independent period, separate from the training phase, is selected to test the performance of downscaling models. Based on the principle of minimum distance, the reanalysis data-driven downscaling models are applied in conjunction with the Monte Carlo random simulation method (Lall and Sharma, 1996) to generate precipitation statistics at monthly and pentad scales. These simulated statistics are then compared against corresponding observational data for validation.

Evaluation Period (2015–2022): During this period, the downscaling models are driven by daily circulation fields derived from the Subseasonal-to-Seasonal (S2S) model at different forecast lead times. Specifically, the circulation data from the dynamic model is projected onto the pre-determined circulation pattern in the training period based on the principle of minimum distance. Then, precipitation is predicted based on the empirical relationship between the circulation pattern and the precipitation probability distribution. Again, Monte Carlo random simulation is employed here to generate the refined spatial structure of precipitation, and the differences between the downscaled and observed precipitation are analyzed to evaluate the model's performance. It should be noted that the effectiveness of the downscaling is influenced by the number of Monte Carlo simulations. When the number is sufficiently large, the results tend to stabilize. Taking into account both

simulation accuracy and computational efficiency, the number of simulations is set to 500 for each station.

3. Establishment of downscaling prediction model

Based on the circulation classification method using a Self-Organizing Map (SOM) neural network, the key circulation patterns influencing early summer monthly precipitation in MLYRB are first identified. Subsequently, the empirical relationships between these circulation patterns and the daily precipitation probability distribution at each station are established (Fig. 3). In Fig. 3a, the columns are numbered from left to right as 1, 2, 3, 4, and the rows are numbered from top to bottom as A, B, C. Based on the column and row numbers, the circulation patterns are named as A1, B1, ..., C4. In Fig. 3b, the cumulative probabilities corresponding to different rainfall amounts can reflect the probabilities of different types of precipitation (such as light rain, moderate rain, heavy rain, etc.). As shown, when the Western Pacific Subtropical High (WPSH) (represented by 5880 gpm isolines over Western Pacific) extends westward, a strong southwesterly flow generally forms on its western flank due to the pressure gradient force. This flow can transport abundant warm and moist air from the ocean toward MLYRB, providing favorable water vapor conditions for precipitation. At this time, the Sea of Okhotsk region shows a strong positive geopotential height anomaly, which is related to the zonal teleconnection structure of the westerly wind belt in the mid-high latitudes, promoting the development of the meridional circulation. The eastern segment of the Meiyu front moves southward, causing the convergence of cold and warm air masses in the areas south of the Yangtze River in China (Wang, 1994; Li et al., 2013). This configuration often leads to heavy rainfall and floods in the MLYRB, thereby increasing the probability of heavy precipitation exceeding 25 mm (e.g., circulation

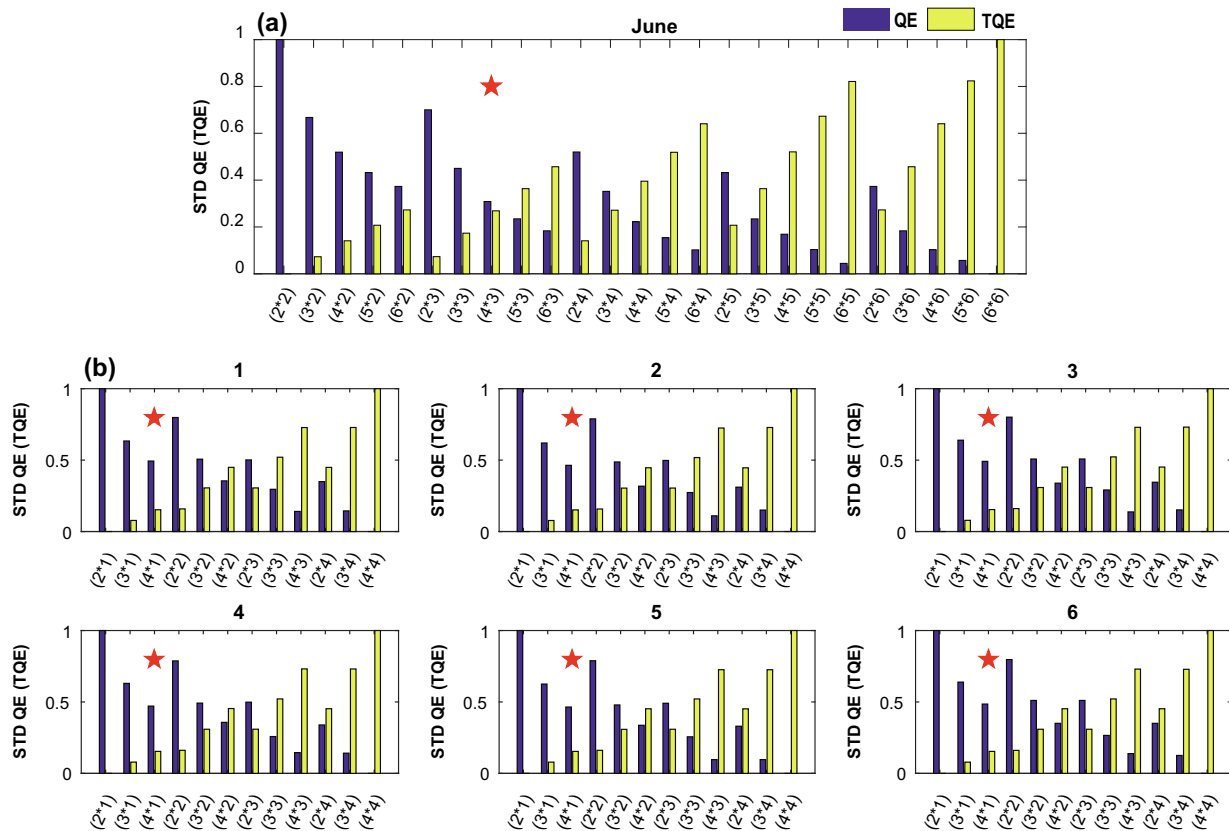


Fig. 2. Classification qualities for different SOM sizes at (a) monthly and (b) pentad scales. Blue and yellow bars represent the standardized quantization errors (QE) and total quantization error (TQE) for each size respectively, and red stars mark classification sizes where both values are relatively low. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

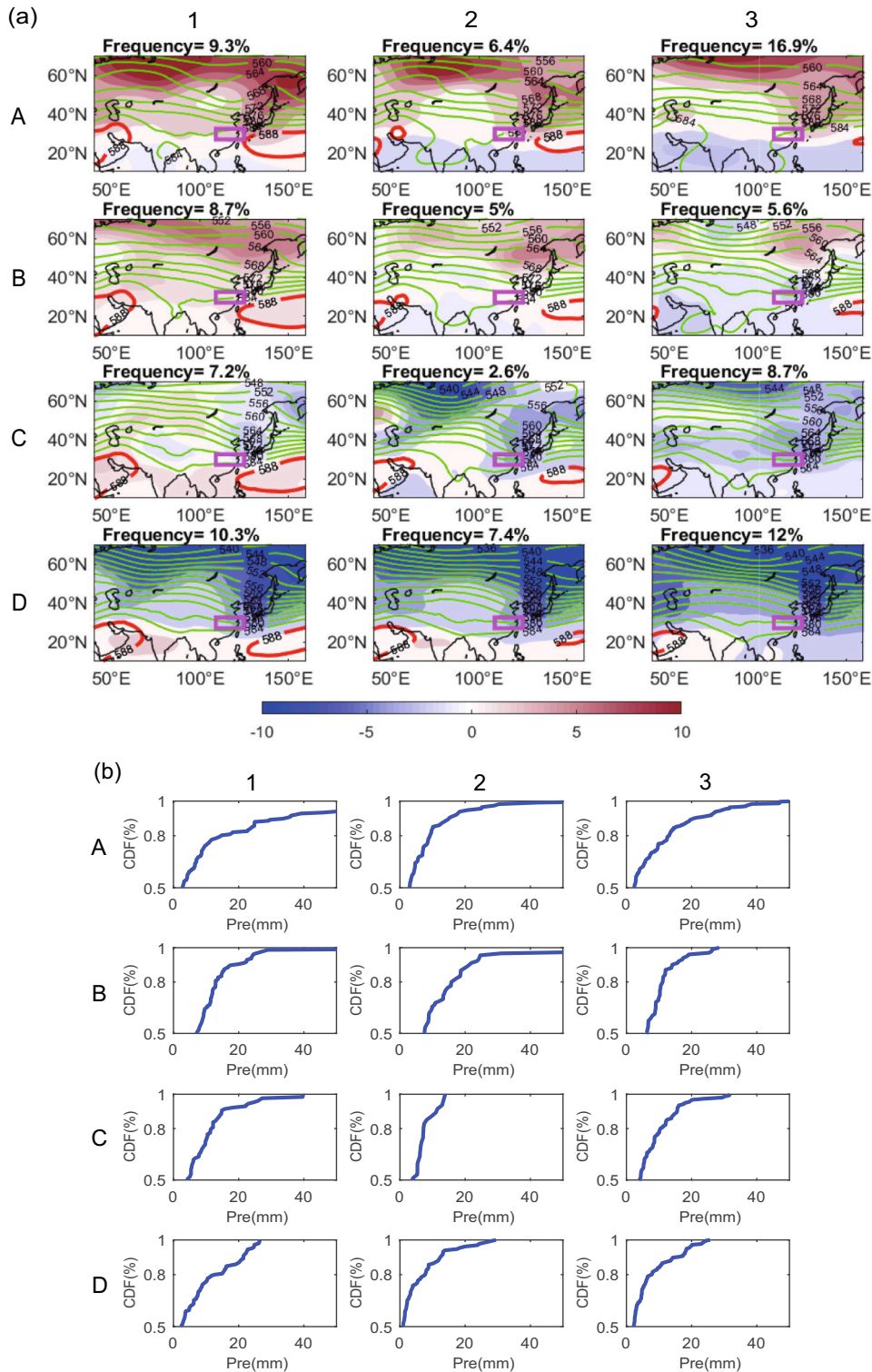


Fig. 3. (a) Classification diagram of key circulation patterns influencing monthly precipitation in the early summer during the training period (1961–1990) (taking Wuhan Station as an example). Green contour lines show the mean 500 hPa geopotential height for each pattern, and the red line indicates the 5880 gpm isolines. The shaded areas represent the anomaly 500 hPa geopotential height of each pattern relative to the early summer climatological mean (unit: dagpm), and percentage values above indicate the frequency contribution of each pattern. The MLYRB area is marked within the purple box. (b) Cumulative probability distribution of precipitation corresponding to each circulation pattern at Wuhan Station, the horizontal axis represents precipitation and the vertical axis represents cumulative density function (CDF). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

patterns A1, B1, C1). Conversely, when the WPSH retreats eastward, the opposite circulation pattern results in a relatively weak upward movement of the air flow in the Yangtze River Basin. Meanwhile, the center of the southwest moist airflow also shifts eastward, causing the MLYRB to

move out of the core water vapor transport zone. Under these conditions, precipitation generally weakens, as shown, the probability of light precipitation below 10 mm increases (e.g., circulation patterns B3, C3, D3).

Furthermore, the relationship between circulation patterns and precipitation probability distributions is also established for each pentad in early summer (Fig. 4). A similar pattern emerges here: heavier precipitation is associated with a westward-displaced WPSH, while lighter precipitation tends to occur when the subtropical high retreats eastward. Additionally, circulation patterns that are adjacent in classification space tend to exhibit similar circulation and precipitation characteristics, whereas those that are more distant show greater variability.

Based on these empirical relationships between circulation patterns and precipitation probability distributions, statistical downscaling models are developed for both monthly and pentad scales to predict early summer precipitation in MLYRB.

Before applying the downscaling method to the simulations from the dynamic climate model, it is essential to independently evaluate its effectiveness. During a time period separate from the training phase, the downscaling models are driven by circulation fields from reanalysis data to assess their abilities to reproduce observed precipitation. For monthly precipitation, the results indicate that the median spatial correlation coefficient between the simulated and observed rainfall in MLYRB is 0.63, reaching as high as 0.84 in certain years. In approximately 75 % of the years, the correlation coefficients pass the 95 % significance level.

For pentad-scale rainfall, there are a few differences in performance across the pentads, the proportion of years in which the correlations between observed and simulated precipitation pass the 95 % significance test for each pentad is 58.3 %, 66.7 %, 75 %, 66.7 %, 66.7 %, and 58.3 %, respectively (Fig. 5). According to the multi-year median spatial correlation coefficients, there is a relatively strong consistency between observed and simulated precipitation for 2–5 pentads with the correlation coefficients ranging from 0.4 to 0.6. This might be due to the fact that the position of rain belt may undergo significant changes during the early and late stages of the early summer, resulting in some differences between the empirical precipitation corresponding to the circulation pattern and the actual situation.

Overall, the downscaling models demonstrate reasonable skill in reproducing the spatial distribution of both monthly and pentad-scale precipitation during most years of the independent validation period.

4. Evaluation of SOM downscaling models

4.1. The performance of downscaling models in simulating the spatial distribution of precipitation

Following the independent test, the realtime outputs of the S2S model are applied to drive downscaling models and evaluate their predictive performances. Firstly, by comparing the pattern correlation coefficients (PCC) between predicted and observed precipitation, the model's ability to simulate the spatial patterns of monthly precipitation in early summer is assessed. As shown in Fig. 6, the correlation coefficients before downscaling are relatively low for all lead times within the two-week forecasting window. Specifically, the multi-year averaged PCCs for lead times of 1–5 days, 6–10 days, and 11–14 days are 0.38, 0.34, and 0.29, respectively. With the advancement of lead time, the prediction skill generally experiences a decline. After downscaling, the spatial prediction accuracy of precipitation improved notably. During the 8-year evaluation period, for different lead times (1–14 days), the proportion of years in which the spatial correlations between down-scaled and observed rainfall reach the 95 % significance level increases from 60 % to 95 %. The multi-year averaged PCCs approximately double compared to the pre-downscaling values, increasing to around 0.6. Moreover, the effect of different lead times within the two-week window on the downscaling performance becomes minimal, implying the promotion of predictable lead times.

The downscaling effectiveness at the pentad scale is also evaluated. By comparing the spatial correlation between predicted and observed rainfall before and after downscaling across different forecast lead times, it is found that the correlations before downscaling are typically below 0.2 and fail to pass the 95 % significance test for most pentads (Fig. 7a). After applying the downscaling approach, the multi-year average spatial correlations pass the 95 % significance test for most pentads. For the 30 lead times (1–30 days), the proportion of years (over the 8-year evaluation period) in which the PCCs between observed and predicted pentad precipitation reach the 95 % significance level increases from 53 % to 92 %. For each lead time within one-month, the PCCs for all six pentad periods show substantial improvements, nearly doubling in most cases. Specifically, the averaged promotions (for different lead time periods) in correlation coefficients for the six pentads are 0.14, 0.20, 0.20, 0.22, 0.23, and 0.31, respectively (Fig. 7b).

Using Taylor diagrams, the long-term predictive performance of downscaling models for monthly and pentad-scale precipitation are

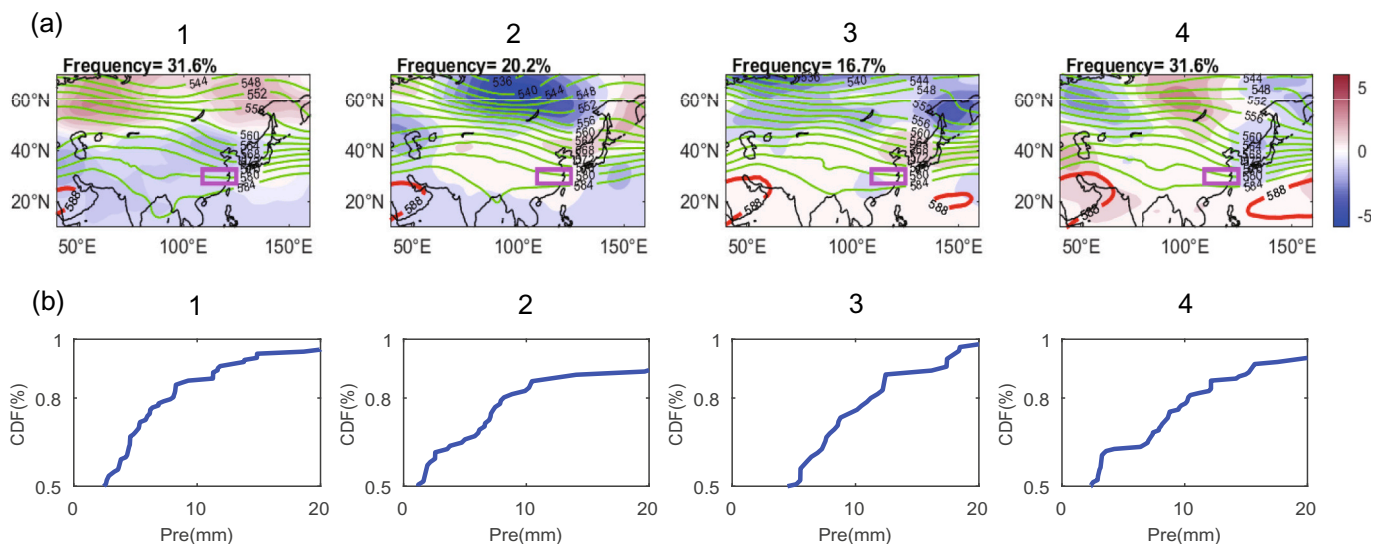


Fig. 4. Same as Fig. 3, but based on circulation classification and cumulative probability distribution of precipitation during the first pentad of early summer at Wuhan Station.

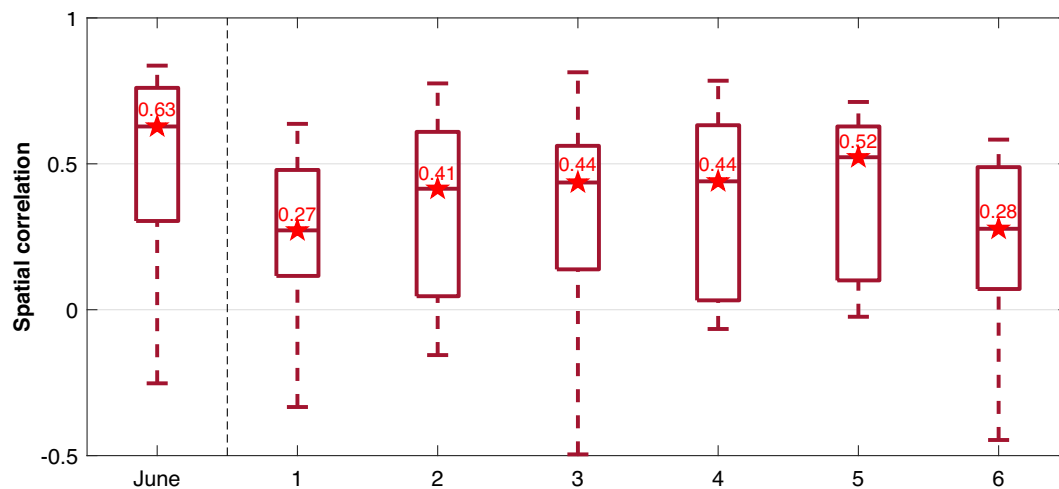


Fig. 5. Boxplots of spatial correlation coefficients between observed and simulated precipitation for early summer (the first boxplot for monthly precipitation, and the following six boxplots for pentad-scale rainfall) during the independent test period (1991–2002). The boxes represent interannual dispersion, and red stars with numbers denote the multi-year median value of correlation coefficients. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

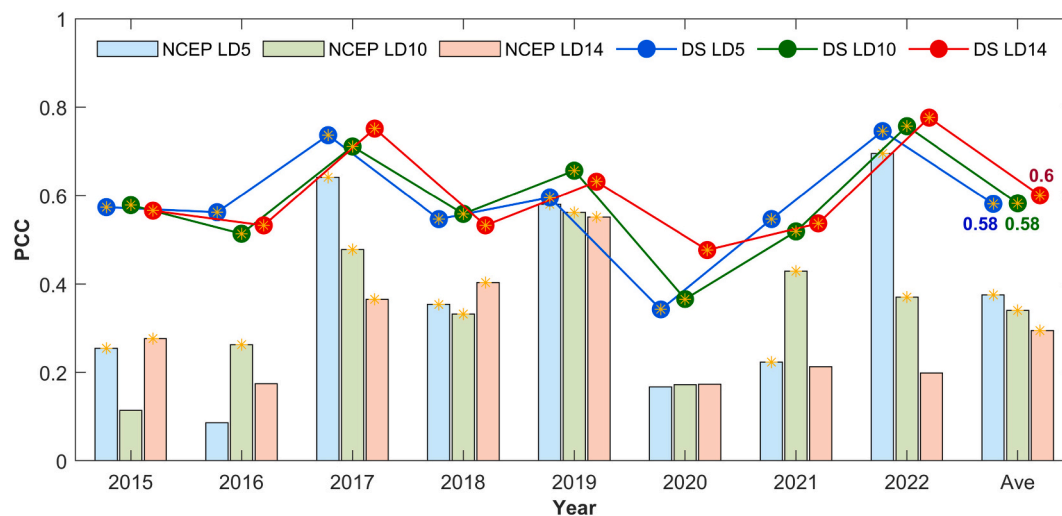


Fig. 6. Pattern correlation coefficients (PCCs) between predicted and observed monthly precipitation during the evaluation period (2015–2022). The columns represent the PCCs before downscaling (NCEP), and the lines represent the results after downscaling (DS). Different colors indicate different lead times (LD): 1–5 days (blue), 6–10 days (green), and 11–14 days (red). The yellow asterisks on the bar charts or line graphs indicate the correlation coefficient passing the 95 % significance test. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

comprehensively evaluated in terms of spatial correlation, normalized standard deviation, and centered root mean square error (centered-RMSE). As shown in Fig. 8, a higher prediction skill is indicated by spatial correlations closer to 1, a standard deviation ratio approaching 1, and a lower centered-RMSE nearing 0. The results reveal that, before downscaling, the spatial correlation coefficients between predicted and observed monthly and pentad precipitation at various forecast lead times mostly range from 0.2 to 0.8. The normalized standard deviations are generally around 0.5 and centered-RMSE values exceed 0.5, suggesting an unsatisfactory performance. After downscaling, for monthly precipitation, the spatial correlations exceed 0.9, the normalized standard deviations approach 1, and RMSE values fall below 0.5. Significant improvements are also found for pentad-scale rainfall. For example, in the fourth pentad, the post-downscaling spatial correlations rise above 0.8, and the normalized standard deviations approach 1, which illustrates a good performance for the SOM downscaling method.

In summary, the SOM downscaling models markedly enhance the prediction accuracy of early summer precipitation in MLYRB at both

monthly and pentad scales. Notably, it elevates the qualities of monthly (pentad) precipitation predictions to a consistent level across different lead times within a two-week (or one-month) window. This improvement in predictable lead time is particularly valuable for enhancing early warning capabilities in this region.

4.2. The performance of downscaling models in capturing the temporal variability of precipitation

In addition to the spatial distribution of precipitation, we also evaluate the predictive capabilities of the downscaling models in capturing the temporal variability of early summer precipitation in MLYRB. For monthly precipitation, using the averaged prediction result with lead times of 11–14 days as an example, before downscaling, the temporal correlation coefficients (TCCs) between observed and predicted precipitation are predominantly negative south of the Yangtze River, with only weak positive correlations to the north (Fig. 9a). After downscaling, the TCC improved markedly, reaching value up to 0.5 north of the

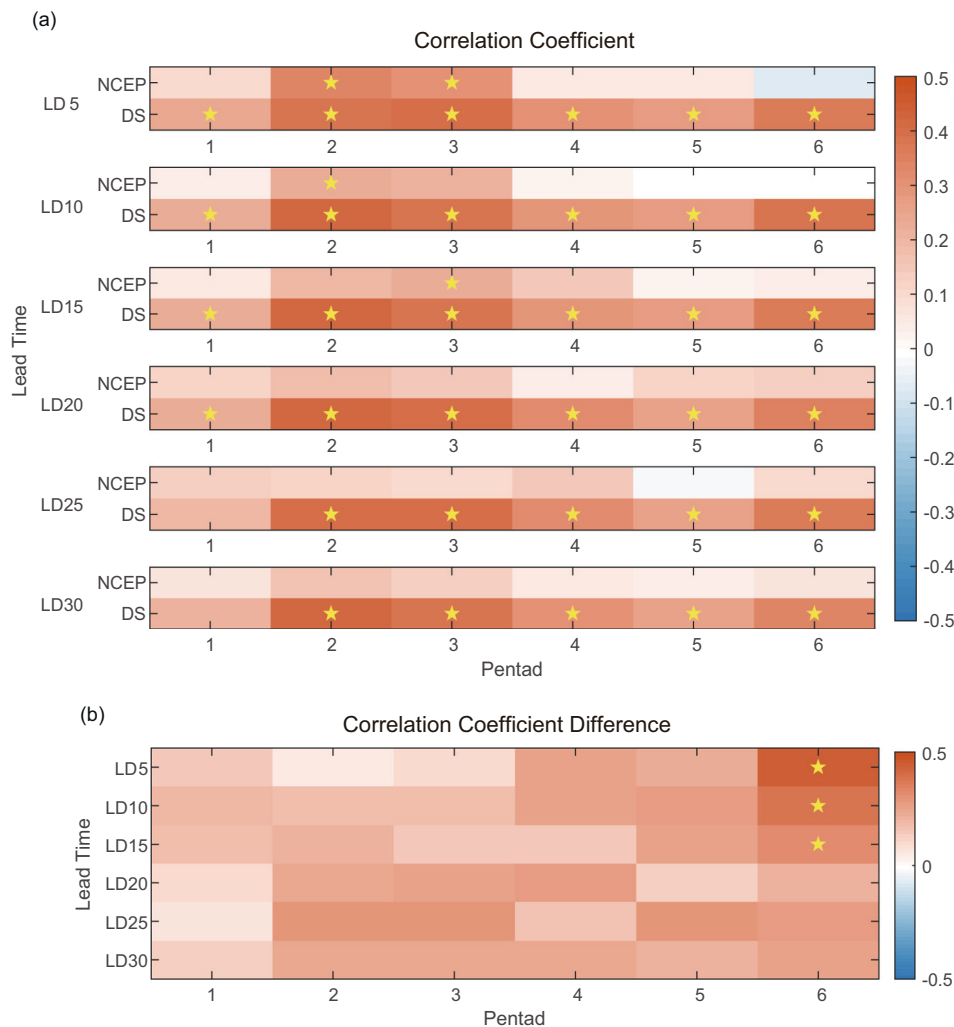


Fig. 7. Pattern correlation coefficients (PCCs) between predicted and observed pentad-scale precipitation during the evaluation period (2015–2022). (a) Multi-year averaged PCCs before (upper row) and after (lower row) downscaling for various lead times (LD): 1–5, 6–10, 11–15, 16–20, 21–25, and 26–30 days. (b) Differences of PCCs before and after downscaling. The yellow pentagrams indicate the correlation coefficient (difference) passing the 95 % significance test. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Yangtze River (Fig. 9b). For pentad precipitation, taking averaged prediction at lead times of 26–30 days as an example, enhancements in the temporal correlations are mainly displayed in south of the Yangtze River after downscaling (Fig. 9c and d).

Despite these improvements, the TCC between observed and predicted precipitation after downscaling still fails to reach statistical significance in most areas of the interested region. This suggests that while the downscaling models enhance the spatial correlation of precipitation predictions, their effectiveness in capturing precipitation anomalies—particularly extreme events—remains limited. In other words, the downscaling approach improves spatial pattern prediction significantly but offers modest gains in predicting the intensity and timing of extreme precipitation in early summer. This might be due to the following reasons: The SOM downscaling method is based on the relationship between circulation patterns and precipitation. And the influence of circulation pattern on precipitation mainly lies in the spatial distribution. As for the intensity of circulation anomaly, the improvement brought by this method may not satisfactory enough. Moreover, the dynamic model itself may be insufficient in depicting the circulation intensity, which together leads to the result that although TCC shows an improvement, it still fails to pass the significance test.

5. Summary and discussion

The early summer rainfall in the middle and lower reaches of the Yangtze River Basin (MLYRB) usually results in cumulative and delayed impacts on livelihoods, production, and the ecological environment. To enhance the accuracy and predictable lead times of subseasonal precipitation predictions during early summer in this region, this study employs a Self-Organizing Map (SOM) neural network-based circulation classification technique to establish a nonlinear relationship between key circulation patterns and the daily precipitation probability distribution. Monthly and pentad-scale statistical downscaling models are developed to capture the fine-scale spatial structure of subseasonal precipitation, and their performances are evaluated by integrating a subseasonal climate dynamic model. The main conclusions are as follows:

Using the SOM method, key circulation patterns influencing monthly and pentad rainfall variability are identified, primarily reflecting the east-west shifts of the Western Pacific Subtropical High (WPSH). A one-to-one correspondence is then established between each circulation pattern and the daily precipitation probability distribution observed in MLYRB during early summer. When the WPSH shifts westward, the probability of heavy precipitation tends to increase, whereas an eastward retreat of the WPSH is associated with light rainfall. Based on this

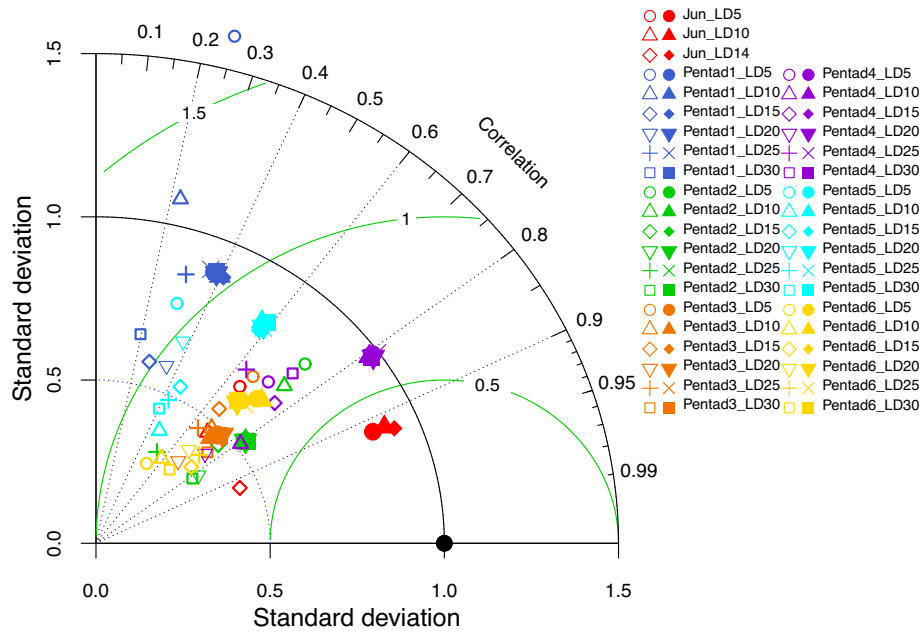


Fig. 8. Taylor diagram showing multi-year (2015–2022) mean early summer precipitation before (hollow) and after (solid) the SOM downscaling. The azimuthal angle represents spatial correlation coefficients, the radial distance from the origin represents normalized standard deviations, and the green arc indicates centered root mean square error. Red colour represents monthly precipitation, different markers indicate different lead times (LD): 1–5 (circle), 6–10 (triangle), and 11–14 days (rhombuse). Other colors represent rainfall in different pentads, different markers correspond to different lead times: 1–5 (circle), 6–10 (triangle), 11–15 (rhombuse), 16–20 (inverted triangle), 21–25 (cross), and 26–30 days (square). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

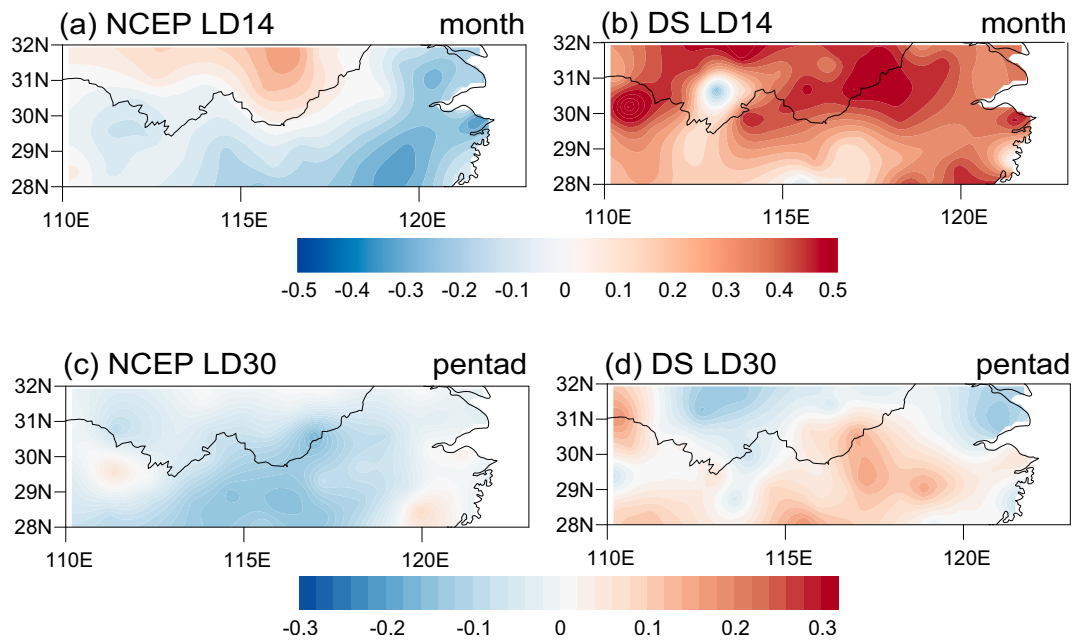


Fig. 9. Spatial distributions of the temporal correlation coefficients (TCCs) of early summer precipitation before and after downscaling. (a, b) Monthly precipitation with a lead time of 11–14 days. (c, d) Pentad precipitation with a lead time of 26–30 days.

empirical relationship between circulation patterns and precipitation distributions, monthly and pentad-scale prediction models are constructed. Independent testing confirmed the model's capability to reproduce the spatial characteristics of observed precipitation patterns.

The outputs of the S2S model are applied to drive downscaling models, and the effectiveness of precipitation prediction is evaluated. Results indicate that, after downscaling, the proportion of years in which the spatial correlations between observed and predicted monthly (pentad-scale) precipitation pass the significance test increases from 60

% (53 %) to 95 % (92 %). The multi-year averaged pattern correlation coefficient also nearly doubles compared to pre-downscaling values, increasing by approximately 0.3 (0.2). In addition to spatial correlation, substantial improvements are also found in the normalized standard deviations and centered root mean square errors for downscaled precipitation. Overall, the statistical downscaling models significantly enhance the ability to predict the spatial distribution of early summer subseasonal precipitation in MLYRB. They also improve the consistency of monthly (pentad-scale) precipitation predictions across different lead

times within two weeks (or one month), illustrating the enhancement of predictable lead times.

Compared with dynamic climate models and traditional statistical downscaling methods, the downscaling approach based on circulation classification demonstrates its advantages in enhancing the spatial simulation capability of subseasonal precipitation patterns and the predictable lead times. This improvement can likely be attributed to the capability of the SOM method to establish nonlinear and physically realistic connections between daily circulation patterns and precipitation (Wilby et al., 2004). Precisely because the SOM-based circulation classification method effectively captures the evolutionary characteristics of key circulation patterns influencing precipitation in the MLYRB, it contributes significantly to enhanced precipitation prediction capability. Nevertheless, its ability to capture precipitation anomalies—particularly extreme events—remains limited. That is, improvements in predicting extreme precipitation are not yet substantial. In fact, whether based on dynamic models, statistical downscaling methods, or AI methods, there is a common challenge in simulating precipitation grades and extreme situations (Guo et al., 2025). For downscaling and AI methods, there are few samples depicting extreme climates in historical observations, making it difficult to capture their characteristics. Therefore, it is still one of the challenges that need to be urgently addressed in the field of climate prediction.

Future efforts should focus on the following aspects: Firstly, the factors influencing regional subseasonal precipitation are complex and variable. In order to further enhance the prediction capability, artificial intelligence (AI) methods such as deep learning can be applied to select the optimal combination of input predictors for the downscaling model. Secondly, due to the strong capability of AI methods in handling nonlinear problems and large datasets, their appropriate application can lead to better performance in precipitation prediction compared to traditional physical-statistical methods (Mayer and Barnes, 2021; Wei et al., 2022; Barnes et al., 2023; Lu et al., 2023). Nevertheless, there is still a lack of clear interpretation regarding the underlying physical mechanisms involved. Therefore, future efforts that combine artificial intelligence with traditional physical-statistical approaches to develop physically constrained AI prediction methods may more effectively enhance subseasonal prediction capabilities. Finally, the SOM method also has the advantage in depicting the characteristics of circulation and precipitation evolution (Li et al., 2024). In the future, it can be utilized to develop downscaling models that can better capture the temporal variability of precipitation and more accurately represent the dynamic evolution of circulation patterns, as well as associated high-impact weather events.

CRedit authorship contribution statement

Mei Li: Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.
Yunyun Liu: Writing – review & editing, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.
Jinqing Zuo: Supervision, Software, Resources, Formal analysis.
Yinghan Sang: Writing – review & editing, Validation, Data curation.
Jiaxi Yang: Writing – review & editing, Resources, Investigation.

Declaration of competing interest

No conflict of interest exists in the submission of this manuscript, and the manuscript is approved by all authors for submission. I would like to declare on behalf of the co-authors that this work is original research which has not been published previously, nor under consideration for publication elsewhere.

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Data availability

Data will be made available on request.

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